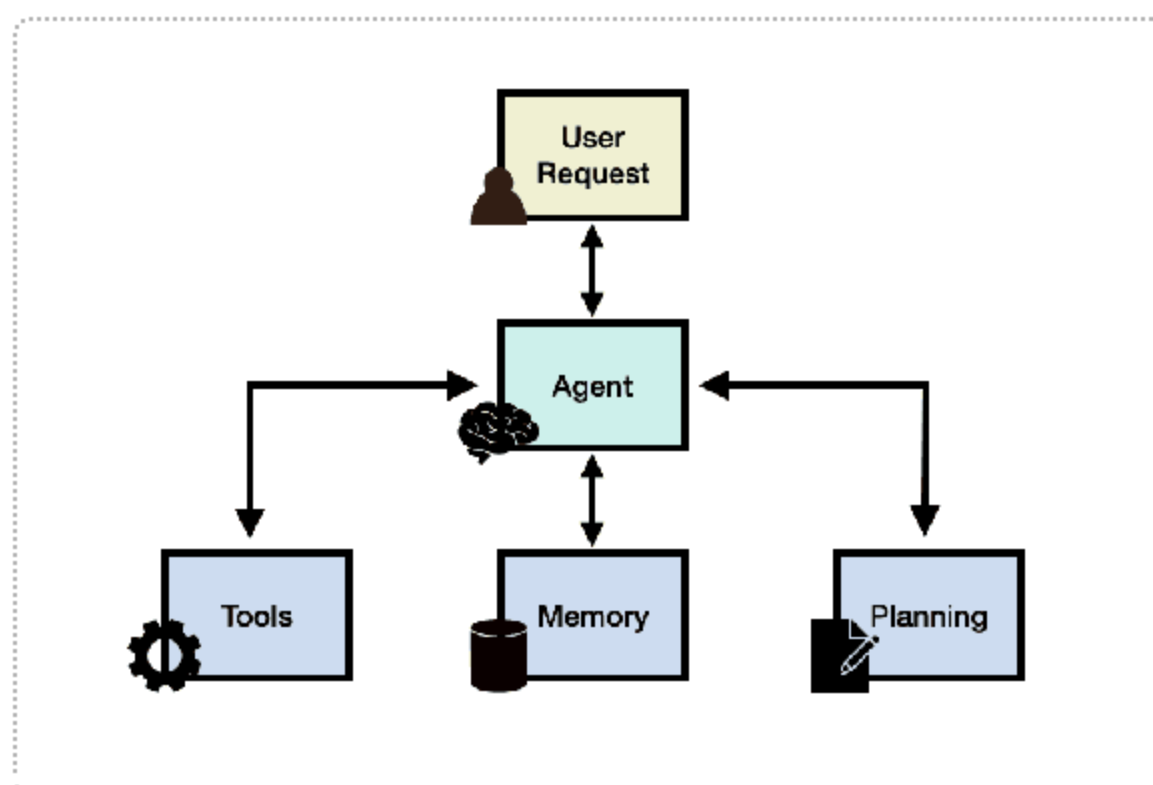


Day 19 of Agentic AI

LLMs and SLMs for Building Agents

Feature	Large Language Models (LLMs)	Small Language Models (SLMs)
Size	More parameters; requires significant computational resources	Fewer parameters; less resource-intensive
Data	Trained on large, diverse datasets; captures broad linguistic knowledge but more susceptible to noise and biases	Trained on smaller, focused datasets; tailored for specific domains, reducing noise and biases
Performance	Excels in a wide range of tasks; versatile and adaptable across domains	Performs well in specialized tasks; high precision in targeted applications
Efficiency	Resource-heavy; higher cost and energy consumption	Highly efficient; runs faster, cheaper, and with lower energy consumption
Use Cases	Suitable for complex, general tasks needing broad language understanding	Ideal for specific, domain-focused applications requiring accuracy and efficiency

How LLMs Power Agents



Large Language Models (LLMs) like GPT-4o and Gemini 1.5 Pro/Flash provide advanced reasoning, text generation, and planning capabilities that empower intelligent agents.

What we'll cover

- **LLMs Overview:** LLMs are AI models trained on vast text data to understand and generate human-like text.
- **Reasoning Capabilities:** They utilize deep learning techniques to analyze information and make decisions based on context.
- **Text Generation:** LLMs predict the next word in a sequence, enabling coherent contextually relevant text.
- **Planning Capability:** They support complex tasks requiring logical reasoning, enhancing automation in various applications.

Key Characteristics of LLMs

Scalability

- LLMs can be scaled to handle vast amounts of data and complex tasks.
- They adapt to various applications, from simple queries to intricate problem-solving.
- **Example:** GPT-4o scales effectively for diverse content generation tasks.

Adaptability

- They can adjust to different contexts and user needs.
- LLMs learn from new data, improving over time.
- **Example:** Gemini 1.5 Pro/Flash adapts to real-time data analysis.

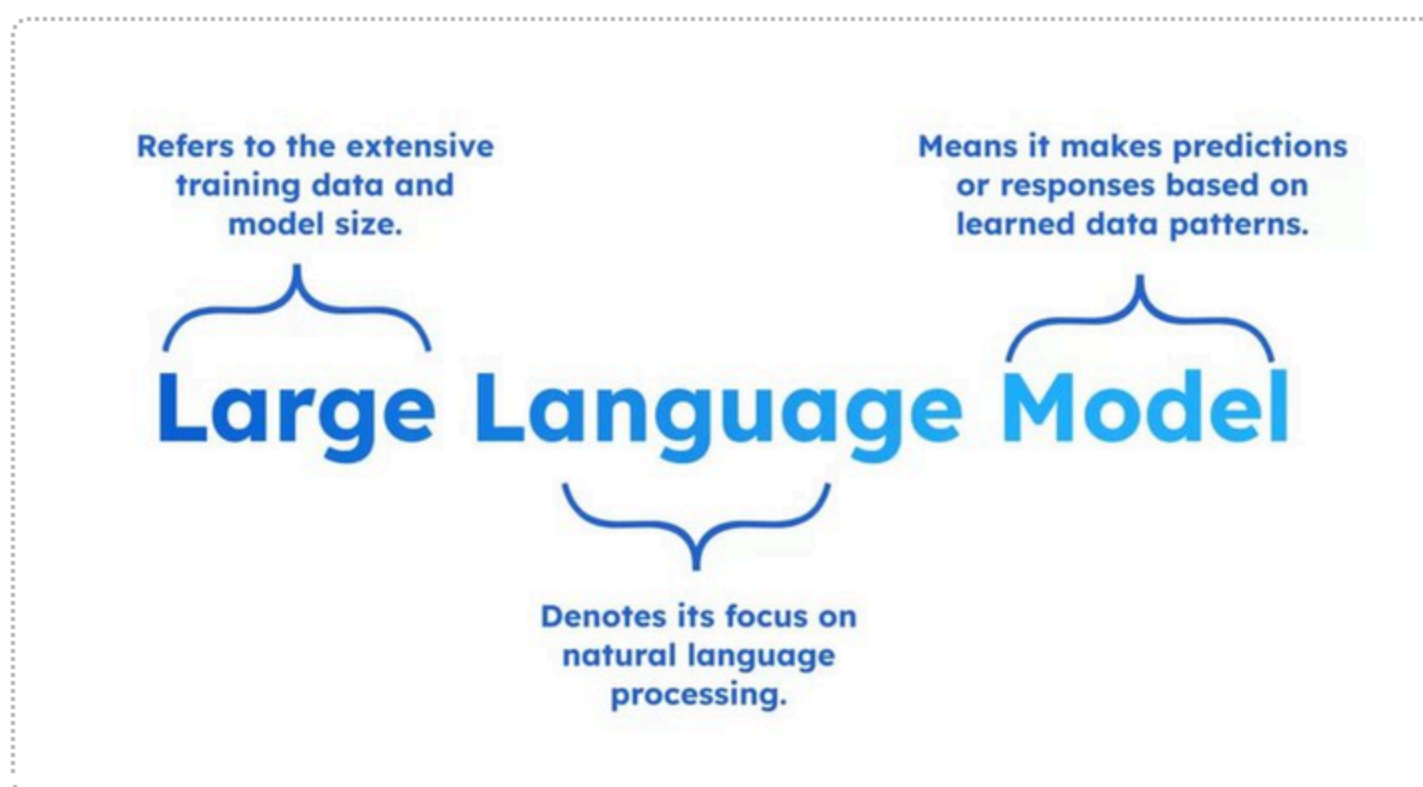
Performance

- High accuracy in generating contextually relevant text.
- They excel in tasks requiring logical reasoning.
- **Example:** LLaMA is used in academic research for natural language understanding.



Understanding LLMs

Large Language Models (LLMs) are AI models trained on extensive text datasets to understand and generate human-like text. They leverage deep learning techniques, particularly transformer architectures, to process language effectively.



How it works

- **Predictive Text Generation:** LLMs generate text by predicting the next word based on context.
- **Contextual Understanding:** They analyze surrounding words to maintain coherence.
- **Application Versatility:** LLMs are employed in various domains, to from chatbots content creation.

Applications of LLMs

Use Case 1

Customer Support

- **Context:** Businesses require efficient customer interaction.
- **Application:** LLMs automate responses to inquiries, improving response times.
- **Benefits:** Enhanced customer satisfaction and reduced workload.

Use Case 2

Content Creation

- **Context:** Content needs to be generated quickly and accurately.
- **Application:** LLMs assist in drafting articles and reports.
- **Benefits:** Increased productivity and creativity in content teams.

Use Case 3

Data Analysis

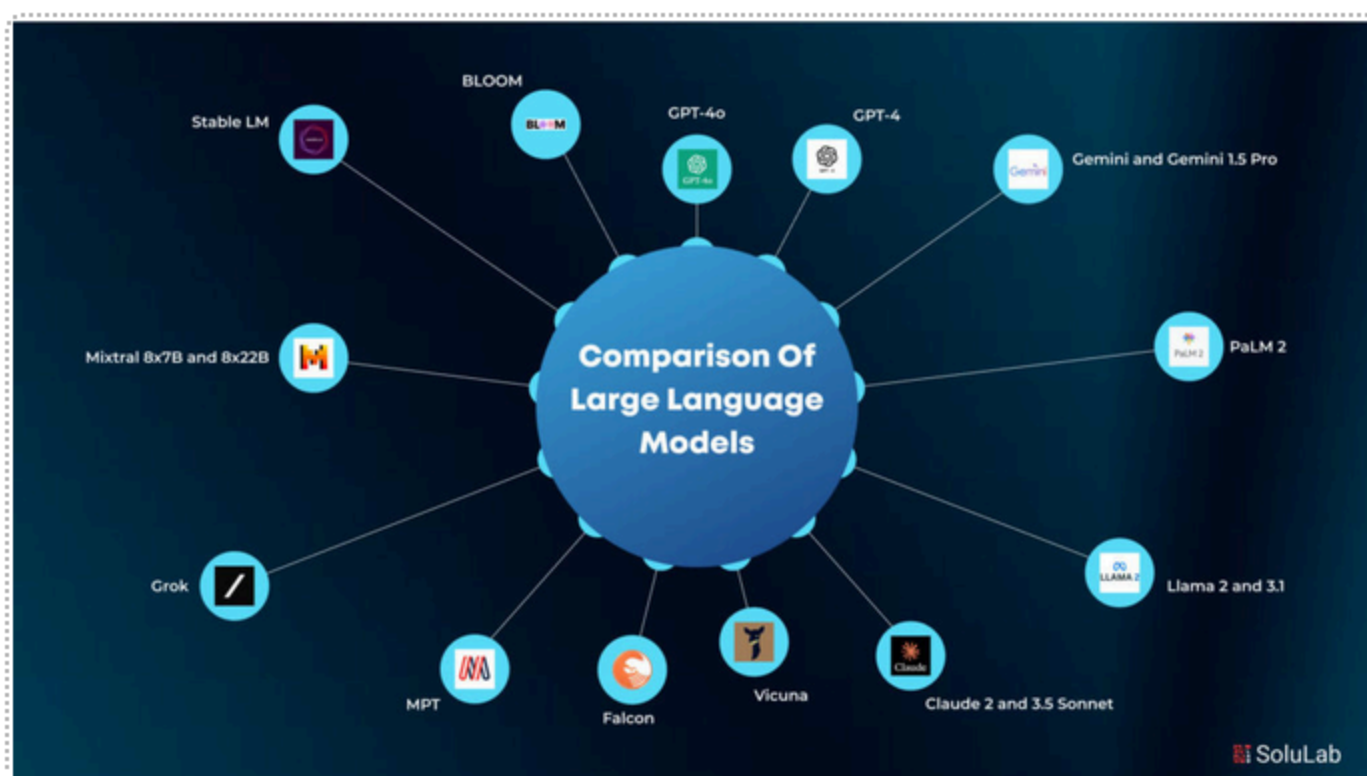
- **Context:** Organizations analyze large datasets for insights.
- **Application:** LLMs process and summarize data effectively.
- **Benefits:** Faster decision-making and improved data-driven strategies.



Characteristics of Modern LLMs

Reasoning-First

- Models prioritize logical reasoning in outputs.
- They analyze information to make informed decisions.
- **Example:** GPT-5 is designed to enhance reasoning capabilities.



Agent-Enabling

- These models facilitate the development of intelligent agents.
- They support complex tasks requiring critical thinking.
- **Example:** Claude 3.7 Sonnet is used in advanced AI assistants.

Performance Metrics

- Modern LLMs are evaluated on accuracy and efficiency.
- They are fine-tuned for specific applications to improve results.
- **Example:** Mistral Magistral is optimized for multi-task learning.

Comparison of LLMs

Option A: LLMs	Option B: SLMs
• High computational requirements	• Lightweight
• Broad generalization capabilities	• Fast processing
• Complex reasoning	• Optimized for edge devices
• Best for: High performance and complex reasoning tasks.	• Best for: Deploying on resource-constrained devices.



Use Cases for Newer LLMs

Use Case 1

Advanced AI Assistants

- **Context:** Users seek personalized assistance.
- **Application:** Newer LLMs like GPT-5 provide tailored responses.
- **Benefits:** Enhanced user experience and satisfaction.

Use Case 2

Complex Problem Solving

- **Context:** Businesses face intricate challenges.
- **Application:** Models like Grok 4 Fast analyze data for solutions.
- **Benefits:** Improved decision-making and efficiency.

Use Case 3

Real-time Data Processing

- **Context:** Organizations need immediate insights.
- **Application:** Models like O4-mini process data in real-time.
- **Benefits:** Faster responses and better operational efficiency.



Characteristics of SLMs

Lightweight

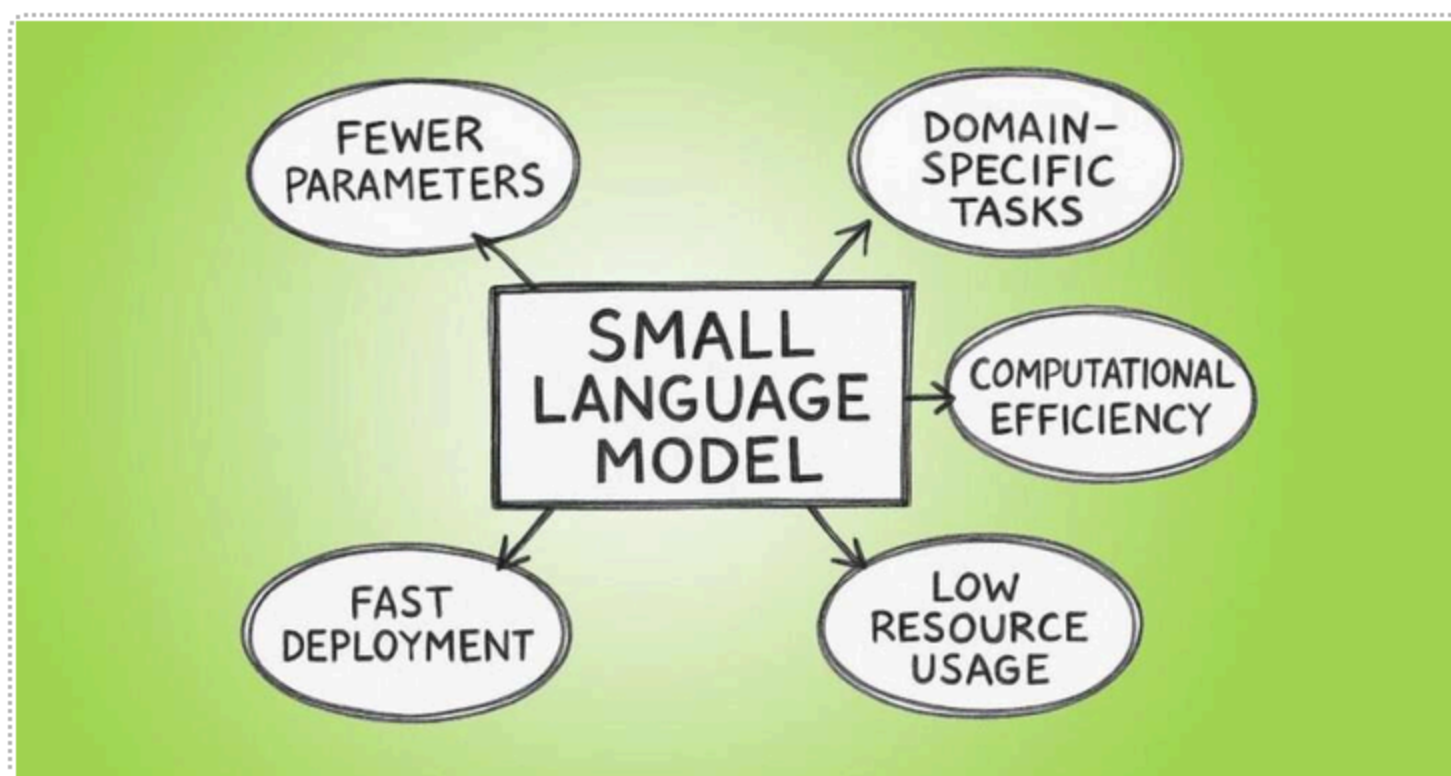
- SLMs are designed to operate with minimal computational resources.
- They are ideal for deployment in edge devices and mobile applications.
- **Example:** Gemma 3 is optimized for smart devices.

Efficiency

- SLMs provide fast inference times, ensuring quick responses.
- They consume less power compared to larger models.
- **Example:** Qwen 3 is used in energy-efficient applications.

Scalability

- SLMs can be scaled for various applications without significant resource demands.
- They adapt to different environments seamlessly.
- **Example:** Phi 3.5 Mini is utilized in diverse IoT scenarios.



Introduction to SLMs

Small Language Models (SLMs) are compact models optimized for deployment on edge devices. They are designed to perform efficiently with limited computational resources.

Importance

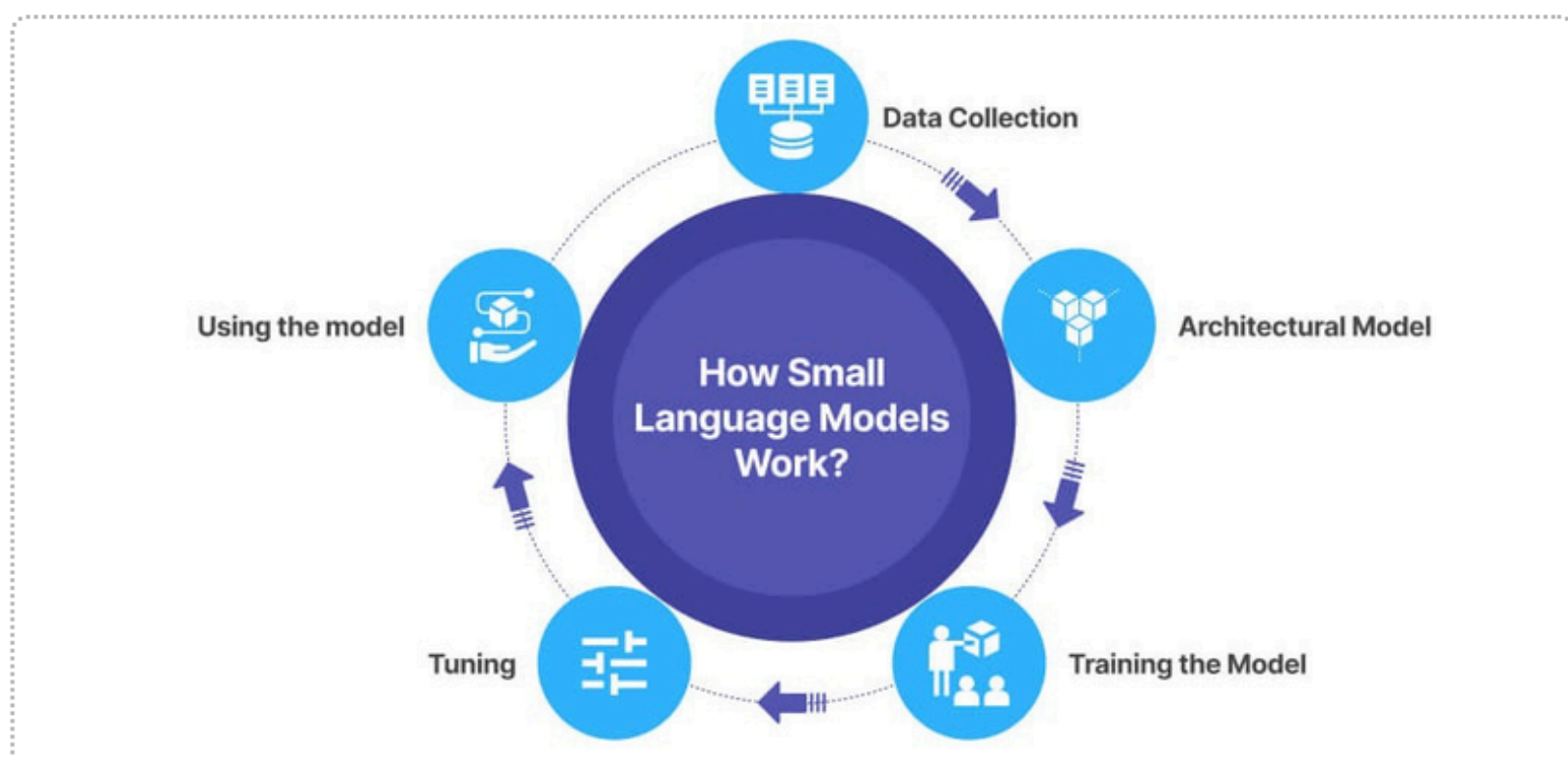
- SLMs enable AI capabilities in environments where computational power is constrained.
- They support lightweight applications across various domains.

Functionality

- **Efficient Processing:** SLMs deliver fast responses with low latency.
- **Resource Optimization:** They are designed to minimize power consumption.
- **Versatile Applications:** SLMs can be adapted for various use cases, including IoT and mobile applications.

Key example

Gemma is implemented in smart home devices for voice recognition, enhancing user interaction.



Use Cases for SLMs

Use Case 1

IoT Devices

- **Context:** Smart home technology requires efficient processing.
- **Application:** SLMs like Gemma power voice recognition in devices.
- **Benefits:** Enhanced user interaction and control.

Use Case 2

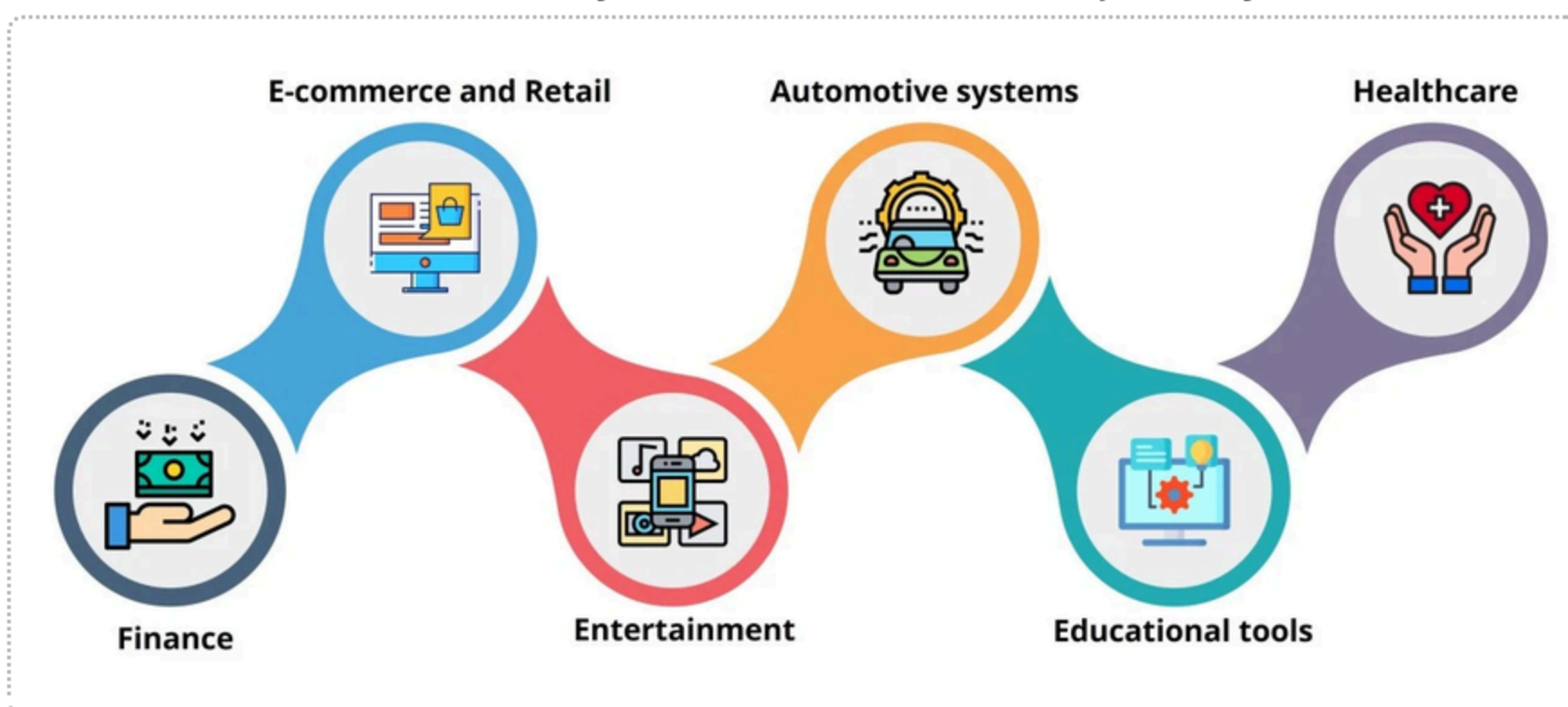
Mobile Applications

- **Context:** Users demand quick responses on mobile platforms.
- **Application:** SLMs are integrated into apps for real-time assistance.
- **Benefits:** Improved user experience and satisfaction.

Use Case 3

Edge Computing

- **Context:** Data processing needs to occur locally for efficiency.
- **Application:** SLMs analyze data on-site without cloud dependency.
- **Benefits:** Reduced latency and increased data privacy.



Key Takeaways

- LLMs and SLMs are crucial in powering intelligent agents.
- Diverse applications exist for LLMs, including customer support and content generation.
- SLMs are increasingly important for lightweight applications on edge devices.
- The choice between LLMs and SLMs depends on specific application requirements.

